

ARTICLE



Efficacy and safety of robotic Cyberknife radiotherapy in uveal melanoma: a systematic review and meta-analysis

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OBJECTIVES: This systematic review is aimed to evaluate the efficacy and safety of robotic Cyberknife radiotherapy (CKRT) in the treatment of uveal melanoma (UM).

METHODS: Clinical studies published in English that assessed the efficacy and safety of robotic CKRT in UM were systematically searched in PubMed, Web of Science, Cochrane and Embase from inception to July 2023. Studies reporting extraocular tumours or other radiosurgery approaches were excluded. Outcomes measured were tumour size, reflectivity, local tumour control rate, eye retention rate, survival rate, complication rate, recurrence rate, and metastasis rate.

RESULTS: Ten eligible articles involving 2370 patients with 2372 UMs were included in evidence synthesis. Meta-analysis showed 811 of 912 patients (0.89, 95% CI: 0.86, 0.92) maintained local control for three years, and 1448 of 1724 patients (0.84, 95% CI: 0.81, 0.88) preserved the eye in three years. During follow-up, 91% (0.91, 95% CI: 0.85, 0.97) patients survived and 351 of 1720 patients (0.23, 95% CI: 0.09, 0.37) involving 1722 eyes had tumour recurrence. In addition, 1376 patients (0.79, 95% CI: 0.77, 0.82) preserved the eyeball in five years. Following treatment, approximately 20% of patients had radiation retinopathy (95% CI: 0.13, 0.28), 19% developed glaucoma (95% CI: 0.11, 0.28), and 22% experienced retinal detachment (95% CI: 0.07, 0.36).

CONCLUSIONS: CKRT is a viable, noninvasive RT for UM. The rate of local tumour control following treatment is comparable to that of other RT approaches, providing additional options based on the patient's condition.

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INTRODUCTION

Uveal melanomas (UMs) are primary intraocular tumours most commonly found in adults [1], especially those over 65 years of age [2]. The incidence of UM is higher in Whites than in Asians [3], which may be attributed to variation in melanin concentration across races. On the other hand, Asian patients exhibit more aggressive clinical symptoms and poorer prognosis [4]. UM is usually located in the choroid (90%) and less often in the ciliary body (6%) and iris (4%) [5]. Despite continual improvement in primary tumour treatments, approximately 50% of UM patients will develop metastases, 90% of which involve the liver [6]. Furthermore, extraocular metastases have been shown to correlate with increased mortality [7].

Several treatment options are available for primary UM, including surgery (enucleation or exenteration), plaque brachytherapy, proton beam radiotherapy, and stereotactic radiotherapy (SRT). Because of the Collaborative Ocular Melanoma study (COMS) [8], treatment for intraocular tumours has shifted toward globe-conserving therapies. Although brachytherapy and proton beam radiotherapy are still the primary treatment modalities, SRT has shown similar efficacy in tumour control and treatment of medium and large UMs.

CyberKnife® (CK®) (Accuray, Sunnyvale, CA, USA) is a highly advanced linear accelerator (LINAC) that performs SRT and fractionated SRT [9]. Compared with conventional radiation therapy (RT), CyberKnife can accurately deliver higher radiation doses to the target while limiting radiation dosage beyond the tumour, facilitating better

preservation of the organs at risk (OAR) [10]. The typical administered radiation dosage is around 20 Gy, which may be applied in either single or fractionated sessions. All patients are positioned supine and immobilized with a thermoplastic mask. Prior to CT and MRI target volume delineation, peribulbar anaesthesia is induced by the injection of lidocaine. To date, several studies have investigated the efficacy of CKRT in UM. However, no meta-analysis has been performed to systematically evaluate the quality, outcomes, complications, and safety of this intervention. Therefore, this study aims to assess the efficacy of cyberknife radiotherapy (CKRT) in local tumour control, complications, reflectivity, secondary enucleations, and overall survival of UM.

MATERIALS AND METHODS

Literature search

This review was conducted as per the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) 2020 statement [11] and registered in PROSPERO (CRD42023451946). Studies published in English that assessed efficacy and safety of robotic CKRT in UM were systematically searched in PubMed, Web of Science, Embase, and Cochrane from inception to July 2023 using the terms “Melanoma”, “Malignant Melanoma”, “Malignant Melanomas”, “Radiosurgery”, “Radiosurgery”, “Cyberknife”, “stereotactic radiotherapy”, “robot”, and “robotic”. Reviews, letters, editorial comments, case reports, conference abstracts, unpublished articles, and non-English articles were excluded. Article selection for this analysis was divided into four steps: identification, abstract and full-text screening,

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eligibility assessment, and inclusion. In addition, two authors completed abstract and full-text screening independently, with any disagreement resolved by consensus.

Eligibility criteria

Inclusion criteria: (1) single-arm study, randomized controlled trial (RCT), case-control study, or cohort study; (2) Adults with UM; (3) CKRT was the intervention; (4) At least one perioperative measure (basal diameter, apical height, reflectivity, complication rate, local tumour control rate, eye retention rate, recurrence rate, and metastasis rate). Two authors independently checked the search results and determined whether inclusion criteria were met by reviewing the titles and abstracts of citations.

Data extraction

Tumour characteristics and patient histories were collected from full texts of each included study. Data were extracted independently by two authors, with any discrepancy resolved by consensus. Data extracted included first author, publication year, study period, study design, tumour size, age, sex, intervention, dose, Collaborative Ocular Melanoma Study (COMS) tumour classification, classification based on American Joint Cancer Committee (AJCC seventh edition), pre- and post-radiotherapy tumour size, reflectivity and visual acuity (VA), complication rate, overall survival (OS), local tumour control, eye retention rate, recurrence rate, and metastasis rate. Local tumour control was defined as tumour size reduction or lack of progression. For continuous variables that were expressed as median and range or interquartile range in the original studies, the mean $\pm$ standard deviation was calculated using a validated mathematical method [12]. When data were missing or not reported in the study, the corresponding authors were contacted to obtain the complete data if available.

Quality assessment

The quality of the included cohort studies was assessed by the Newcastle–Ottawa Scale (NOS) [13], and studies with seven to nine points were regarded as high quality. The quality of single-arm studies was evaluated using the Joanna Briggs Institute (JBI) critical appraisal tool. The quality and level of evidence of eligible studies were evaluated independently by two investigators, and any discrepancy was resolved through consensus.

Statistical analysis

Evidence synthesis was performed in Review Manager v5.3 (Cochrane Collaboration, Oxford, UK). Continuous and dichotomous variables were compared and reported as WMD (weighted mean difference) and OR (odds ratio) plus 95% confidence interval (CI), respectively. Heterogeneity was determined by the X^2 test (Cochran's Q) and I^2 index [14]. A Cochran's Q test p value > 0.1 and $I^2 < 50\%$ indicate no or low heterogeneity, and a random-effects model was used for meta-analysis; otherwise, a fixed-effect model was utilized. In addition, one-way sensitivity analyses were performed to determine the impact individual studies have on the pooled results. Publication bias in outcomes that were reported in three or more studies was evaluated by a funnel plot in Review Manager v5.3 and Egger's regression test [15] in Stata 12.0 (Stata Corp, College Station, TX, USA). A $p < 0.05$ was considered as statistically significant.

RESULTS

Literature search and study characteristics

Our initial search yielded 616 studies from PubMed, Web of Science, Cochrane and Embase, and 509 articles were screened after removal of duplications (Supplementary material 1). Abstract screening excluded studies that were not relevant to the topic, reviews, case reports, conferences, comments, and non-English studies. Full-text screening yielded 14 potentially eligible studies, four of which were excluded due to the application of other SRT or non-CKRT primary therapy, or without sufficient data. A final total of ten full-text [16–25] studies involving 2370 patients (2372 eyes) were included for meta-analysis. There were one prospective single-arm study [16], eight retrospective single-arm studies [17–24], and one retrospective cohort study [25].

Among the patients included in the analysis, 47% were women. The baseline characteristics of all eligible studies are summarized in Table 1. A summary of the site of radiation dose, follow-up,

Table 1. Baseline characteristics and methodological assessment of include studies.

Author	Study period	Country	Study design	Patients	age	Female (%)	Follow-up (months)	CKRS Dose (Gy)	COMS classification (%)			AJCC (%)			T1–T7			Initial VA	Final VA
									Small	Medium	Large	T1	T2	T3	T4	T5	T6		
Choi et al. [24]	2004–2007	Korea	Retrospective	6	26.83 $\pm$ 6.57	50.	26.83 $\pm$ 6.57	36 and 39	0	66.7	33.3	NR	NR	NR	NR	NR	NR	0.42 $\pm$ 0.15	HM(1);NLP(3);0.6(1);0.4(1)
Eibl-Ludhne et al. [16]	2016	Germany	Prospective	217	64 (21–95)	43	29.6 (5.9–84.0)	20.3 (17–22)	3.31	66.9	29.8	NR	NR	NR	NR	NR	NR	NR	NR
Yazici et al. [17]	2007–2013	Turkey	Retrospective	182	54 (18–82)	43	24 (2–79); 7 were lost	<45 Gy (39%); $\geq$ 45 Gy (61%)	5.5	49.5	49.5	13	20	38	8	21		1.0 (26.2%) : partial vision(63.1%);loss of vision(10.7%)	66%decreased; 33% stable; 1%increased
Özcan et al. [19]	2014–2019	Turkey	Retrospective	36	60.5 $\pm$ 14.8 (28–86)	36.	17.2 $\pm$ 10.43 (6–48)	21	NR	31	3	11.1	27.8	52.8	8.3	0		0.5 $\pm$ 0.3 logMAR	0.6 $\pm$ 0.3 logMAR
Liegl et al. [18]	2005–2018	Germany	Retrospective	101	67 (26–85)	44	31.5 (6–133)	17–21	9.1	NR	NR	6.9	20.1	62.4	7.9	3.7		NR	NR
Siedlecki et al. [25]	2007–2013	Germany	Retrospective	268	62.5 (20–90)	48	54.70 $\pm$ 34.20 (13.3–153.6)	Median 20	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR	NR
Yazici et al. [21]	2007–2019	Turkey	retrospective	445	56 (18–80)	48	74 (1–173)	20 (8–22)	0	70	30	20	36	34	5	5		NR	NR
Liegl et al. [22]	2005–2019	Germany	Retrospective	594	64.5 (20.9–94.5)	49	35.1 (0.4–156.7)	21 (17–22)	NR	70	20.9	22.7	57.9	18.9	0.5	0		NR	NR
Schmelter et al. [23]	2005–2019	Germany	Retrospective	74	64 (30–82)	57	54.70 $\pm$ 34.20 (13.3–153.6)	20.3 (18–22)	NR	NR	NR	16	31	21	6	0		NR	NR
Badran et al. [20]	2009–2013	Turkey	retrospective	29	56 (27–75)	41.	At least 15	50 (12–62.5)	0	86.2	23.8	NR	NR	NR	NR	NR	NR	NR	NR

NR not reported.

tumour size, tumour classification, and basic patient information is also provided in the table.

Tumour size

Pooled analysis of basal diameter from four studies involving 901 patients [16, 19, 22, 24] revealed no significant difference after CKRT (SMD: -0.29 ; 95% CI: $-0.73, 0.15$; $p = 0.20$). There was significant heterogeneity among studies ($I^2 = 80\%$, $p = 0.002$) (Fig. 1), and sensitivity analysis showed that the results were not robust. However, there was no significant publication bias as indicated by the Egger's test ($p = 0.43$) (Fig. 2).

Another analysis of apical height from five studies with 434 patients [16, 18, 19, 22, 24] indicated significant reduction after radiotherapy (MD: -1.90 ; 95% CI: $-2.81, -0.99$; $p < 0.0001$). There was significant heterogeneity among studies ($I^2 = 83\%$, $p = 0.0001$) (Fig. 1) but sensitivity analysis indicated that the results were robust. In addition, the Egger's test ($p = 0.43$) did not reveal any significant publication bias (Fig. 2).

Reflectivity

Three studies involving 392 patients were included in the analysis [16, 18, 23]. Pooled results demonstrated that the reflectivity, a marker of fibrosis, improved significantly after treatment (MD: 17.42 ; 95% CI: $15.95, 18.89$; $p < 0.00001$) without significant heterogeneity among studies ($I^2 = 0\%$, $p = 0.79$) (Fig. 1). Sensitivity analysis showed that the results were reliable. Additionally, no significant publication bias was identified by the Egger's test ($p = 0.28$) (Fig. 2).

Local tumour control

Meta-analysis of four studies involving 1355 patients [16, 18, 21, 22] indicated that 811 of 912 patients (0.89, 95% CI: $0.86, 0.92$) had local tumour control (regression or no growth) at three years (Fig. 3), and

1057 of 1355 patients (0.78, 95% CI: $0.73, 0.83$) (Fig. 3) had local tumour control at 5 years post treatment.

Eye retention rate

Six studies reported eye retention rate in 1729 patients [16–19, 22, 23]. Evidence synthesis showed that 1376 patients (0.79, 95% CI: $0.77, 0.82$) eventually preserved the eyeball at five years post-treatment (Fig. 3).

Survival rate

Six studies with 1477 patients (1480 eyes) [16, 17, 19, 21, 22, 24] were included in the analysis for OS. Pooled analysis indicated that 91% of patients (0.91, 95% CI: $0.85, 0.97$) (Fig. 3) survived in the follow-up period. In addition, four studies reported involving 737 patients [18, 19, 22, 24] disease-specific survival rate. Meta-analysis indicated that 92% of patients (95% CI: $0.87, 0.97$) survived, excluding non-disease-related causes of death (Fig. 3).

Complications

All ten studies reported the type and number of CKRT-induced complications. The most common complications are radiation-induced retinopathy and glaucoma. Meta-analysis of nine studies with 2435 patients (2438 eyes) [16–19, 21–25] showed that 537 patients (0.20, 95% CI: $0.13, 0.28$) developed radiation-induced retinopathy (Fig. 3). Furthermore, glaucoma was observed in 340 of 1892 patients (1893 eyes; 0.19 , 95% CI: $0.11, 0.28$) (Fig. 3) [16–19, 22–25].

Cataract was observed in 177 of 798 patients (799 eyes) [17–21, 24] from six studies (0.29, 95% CI: $0.19, 0.39$) (Fig. 3), and retinal detachment was reported in 452 of 1660 patients (1662 eyes) from six studies (0.22, 95% CI: $0.07, 0.36$) (Fig. 4) [16, 17, 20, 22–24]. Maculopathy developed in 36 of 288 patients from four studies (0.32, 95% CI: $0.04, 0.60$) (Fig. 4) [16, 19, 20, 24].

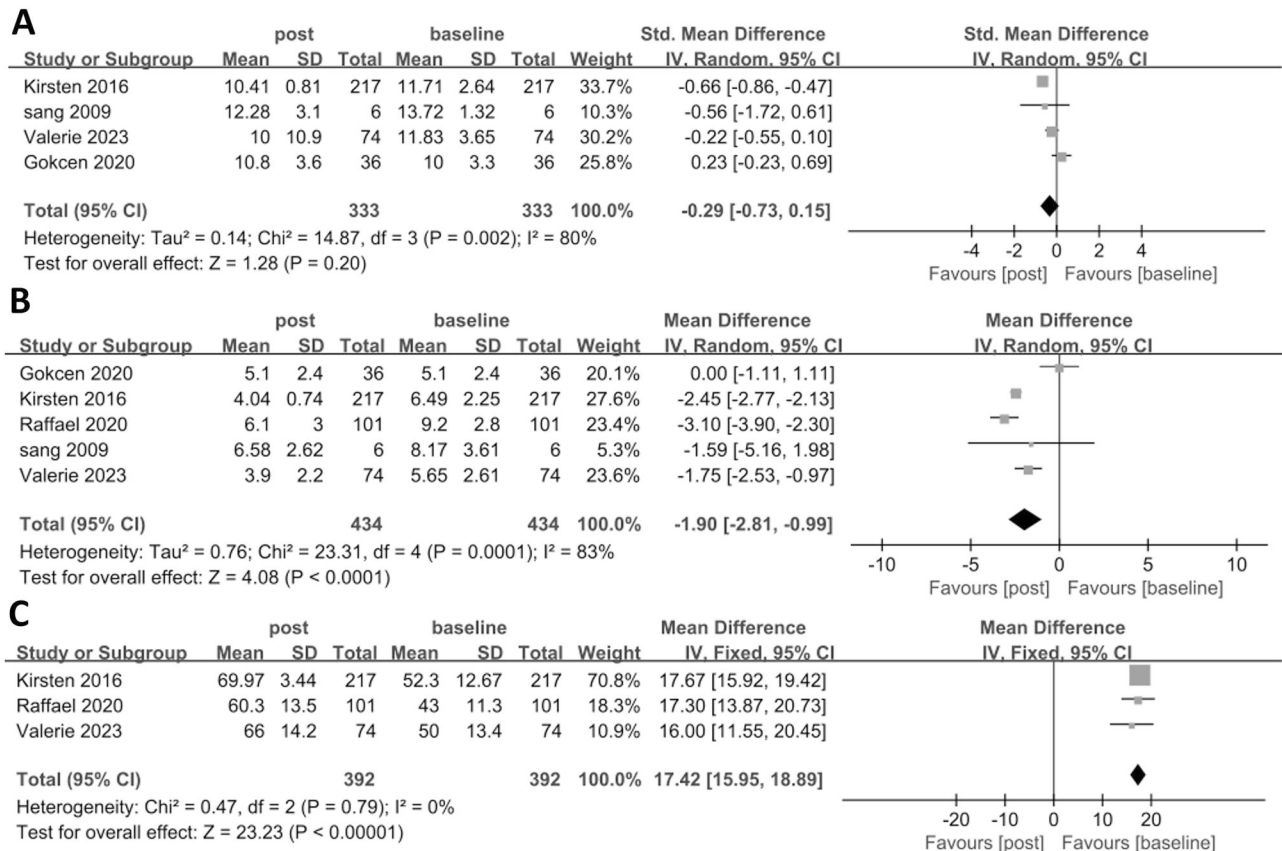


Fig. 1 Forest plots of oncological outcomes. A Basal diameter, B Apical height, C Reflectivity.

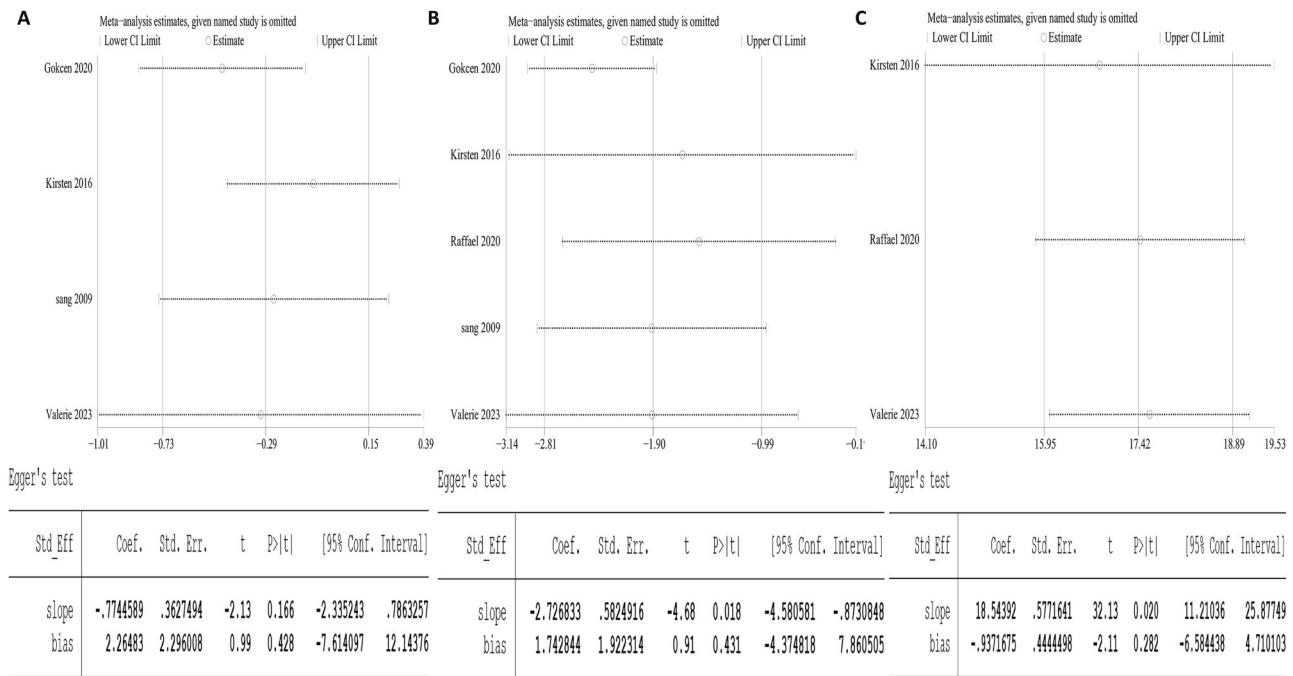


Fig. 2 Sensitivity analysis results. A Basal diameter, B Apical height, C Reflectivity.

Vitreous haemorrhage was detected in 115 of 1022 patients (1023 eyes) from five studies (0.11, 95% CI: 0.05, 0.16) (Fig. 4) [16, 17, 20, 22, 24].

Recurrence rate

Recurrence rate was pooled from six studies involving 1720 patients (1722 eyes) [16–19, 22, 23]. Pooled analysis revealed that 351 patients (0.23, 95% CI: 0.09, 0.37) had tumour recurrence during follow-up (Fig. 4). Some of the patients who relapsed required enucleation (0.12, 95% CI: 0.09, 0.16) (Fig. 4) or repeat radiosurgery (0.04, 95%CI: 0.03, 0.05) (Fig. 4).

Metastasis rate

Three studies reported metastasis rate in 810 patients (811 eyes) [17, 19, 23]. Pooled analysis indicated that 16% patients (95% CI: 0.03, 0.29) (Fig. 4) developed distant metastases during follow-up.

DISCUSSION

Primary choroidal melanoma can be treated by various approaches, ranging from local resection and primary enucleation to RT. RT not only achieves high disease control rate but also preserves the eye and maintains vision. Brachytherapy was once the most commonly used RT, but its limitations on height and transverse diameter renders this treatment unsuitable for medium and large tumours [26]. In recent years, several treatment options have gained popularity, such as proton beam irradiation, gamma knife radiotherapy, and CKRT [27–29]. However, proton beam RT is unavailable in many countries, particularly in those with a low incidence of UM. Therefore, SRT has become a viable alternative. In addition to its low cost, SRT utilizes noninvasive fixation systems to deliver the radiation dose [24, 30]. In the CKRT system, target localization, image acquisition and alignment corrections are repeated every 30–60 s during treatment delivery, which not only increases the efficiency and accuracy of the procedure but also minimizes the treated and irradiated volumes, thereby achieving high-dose conformity and steep dose gradients in all directions [31].

RT is the preferred treatment for most choroidal melanomas, but it also carries with a number of inevitable complications, some of which are seriously sight-threatening. Therefore, a comparative

analysis of the complications associated with the various radiation therapies is required. First, radiation causes various adverse consequences, including radiation retinopathy, macular degeneration, retinal pigment epithelium fibrosis, atrophy, and hyperplasia, as well as alterations in the retinal vascular endothelium, leading to pathological changes such as retinal ischemia, capillary dilation, and microaneurysms [32]. Guleser et al. [33] examined the prognosis of patients receiving plaque brachytherapy and found the development of radiation maculopathy (RM) in 23.4% of patients and radiation retinopathy (RR) in 39.3% of patients. Our study revealed that 20.0% (95% CI: 0.13, 0.28) of patients developed radiation retinopathy following CKRT, whereas 32% (95% CI: 0.04, 0.60) developed maculopathy. The decreased incidences of these radiation complications in our study may be attributed to the use of a lower radiation dose and increased precision in radiation application. Furthermore, radiation doses greater than 10 Gy can induce lens fibre distortion, resulting in cataract development [34]. Cataracts are common following RT. Research has shown that 35% of patients receiving plaque brachytherapy and 20–62% of those receiving proton beam radiation acquired cataracts [35]. Consistent with these findings, our data also indicated that 29% of patients (95% CI: 0.19, 0.39) acquired cataracts following CKRT. Secondary glaucoma is also a common consequence of radiotherapy in UM, which may be caused by either the tumour or the radiation. Radiation can induce glaucoma by inducing angle fibrosis or angiogenesis. Gollrad et al. [36] reported that 21% of patients had secondary glaucoma after proton beam radiation. Similarly, we found that 19% of patients (95% CI: 0.11, 0.28) experienced secondary glaucoma following CKRT. Furthermore, some individuals developed vitreous bleeding following radiotherapy, which may be attributed to the rupture of new vessels caused by radiation retinopathy or the rupture of retinal or tumour vessels caused by retinal tumour invasion [37]. This observation was more prevalent in patients with tumours larger than 5 mm [38]. In the present study, 11% of patients (95% CI: 0.05, 0.16) developed vitreous haemorrhage after undergoing CKRT. van Beek et al. [39] demonstrated that the incidence of vitreous haemorrhage was lower in proton beam radiation (3.3%) than in fractionated SRT (18.3%). Exudative retinal detachment has been reported in UM, but it generally resolves following RT [40]. However, some patients may still experience retinal detachment after the

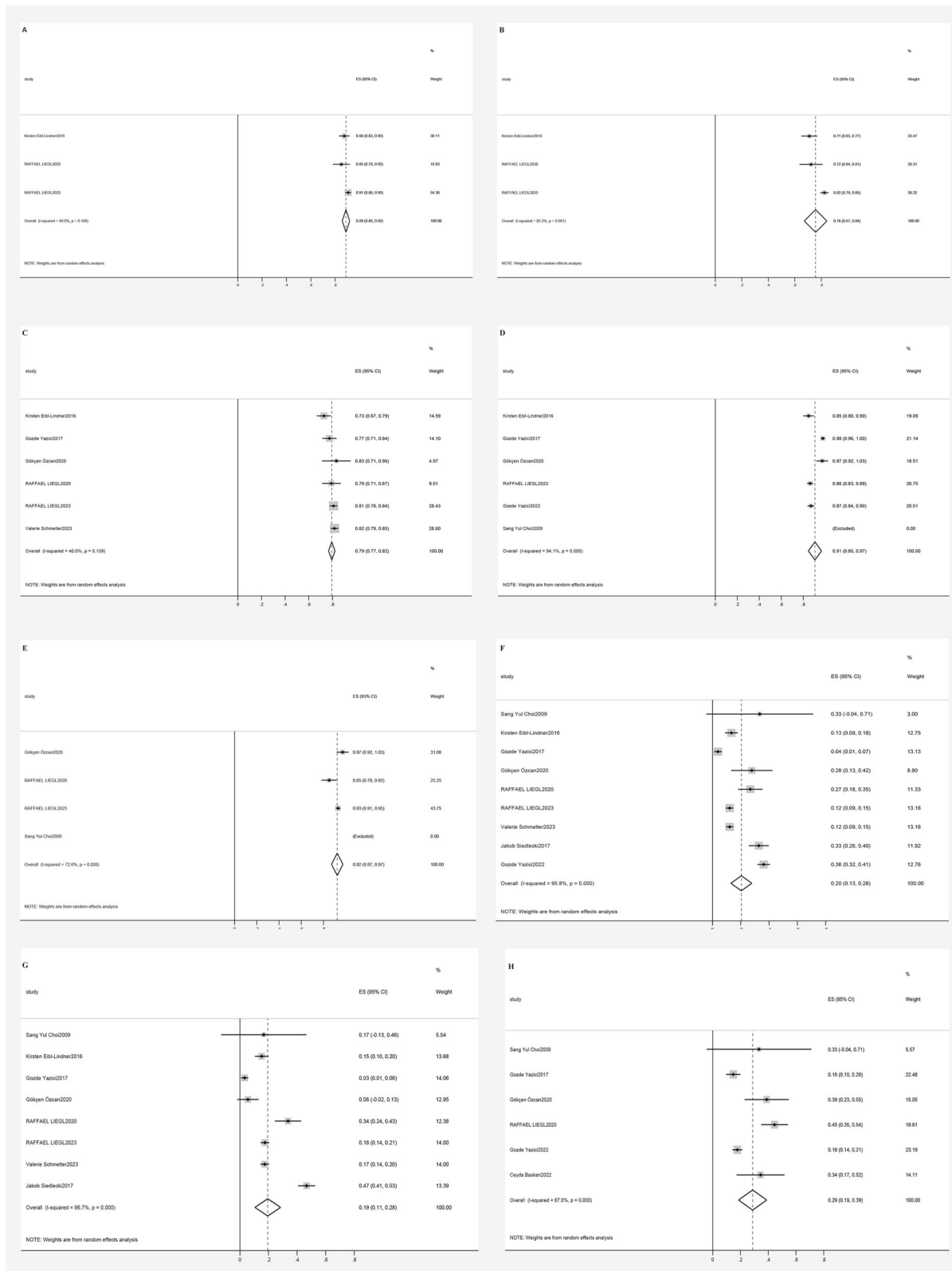


Fig. 3 Forest plots of pooled ratios for each outcome. A Proportion of patients who had local tumour control in 3 years, **B** Proportion of patients who had local tumour control in 5 years, **C** Proportion of patients who preserved the eyeball at 5 years, **D** Proportion of patients who survived in the follow-up period, **E** Proportion of patients who survived excluding non-disease-related causes of death, **F** Proportion of patients who developed radiation-induced retinopathy, **G** Proportion of patients who developed glaucoma, **H** Proportion of patients who developed cataract.



Fig. 4 Forest plots of outcomes. A Proportion of patients who developed retinal detachment, **B** Proportion of patients who developed maculopathy, **C** Proportion of patients who developed vitreous haemorrhage, **D** Proportion of patients who had tumour recurrence, **E** repeat enucleation, **F** repeat radiosurgery, **G** metastasis.

completion of treatment. In such cases, retinal detachment may be associated with retinal thinning, atrophy, or atrophic retinal tears. In our patient cohort, 22% of patients (95% CI: 0.07, 0.36) experienced retinal detachment after undergoing CKRT.

Moreover, tumour recurrence rate remained high (0.23, 95% CI: 0.09, 0.37) following CKRT. This may result from inadequate radiation exposure at the tumour margin, allowing tumour cells to spread via exudative retinal detachment and form tumour tissue in the

peripheral retina. Radiation resistance can lead to the recurrence of late-stage cancers. Vision impairment is another significant concern after RT, as it often leads to poor visual outcomes. This is closely associated with radioactive fundus lesions, radiation dosage, tumour infiltration, and tumour distance from the macula and optic nerve. In this study, over 50% of patients undergoing CKRT reported severe visual impairment (Table 1), which is in agreement with a previous clinical assessment [41] demonstrating that 61% of patients had serious eyesight degradation or loss following gamma knife radiation.

This is the first meta-analysis of CKRT in the treatment of UM. However, due to several limitations, our results should be interpreted with caution. First, one continuous variable (basal diameter) had high heterogeneity and sensitivity analysis showed that the results were not robust, only for statistical description. Second, our study was retrospective in nature and had incomplete data on patient and tumour features and treatment outcomes. Therefore, we were unable to perform subgroup analysis. Nevertheless, subgroup analyses according to tumour size, treatment regimen, and dose should be included in future analysis. Third, most of the included studies were single-arm studies, with no parallel comparisons, which may introduce a bias factor CKRT has several advantages over other treatments such as widespread availability, higher dose conformity, lower cost and minimal invasiveness. Despite favourable outcomes, toxicity remains a major limitation of this treatment. Our analysis indicated that the incidence of CKRT-related complications is relatively high and most complications originated from radiation. Some studies have noted that the incidence of complications is related to radiation dosage and tumour distance from the optic nerve [10, 42, 43]. Dunavoelgyi et al. [10, 42, 43] reported that the incidence of complications increased as the tumour approaches the optic disc and macula and when radiation was administered at >50 Gy in five fractions at 80% isodose. Espensen et al. [43] reported dose-response complications after proton RT. The authors found that optic disc $D_{20\%}$ and cornea $D_{20\%}$ had the greatest impact on optic neuropathy and neovascular glaucoma, respectively, and retina $V52Gy$ correlated with the risk of retinal detachment. These results indicate that dose optimization is important for reducing complications and is hence a key direction of research in future studies. The study by Gozde et al. [21] indicated that treatment of the eye every other day can minimize toxicity and enucleation. This conclusion also provided new ideas for future treatment options. For example, intraocular injection has become the mainstream treatment and preventive measure for radiation-induced intraocular complication. Intravitreal bevacizumab is currently the most common treatment [44], and many drug combinations and new drugs are being tested [45–50].

CONCLUSION

CKRT is an effective, noninvasive treatment for improving tumour local control rate and eye preservation rate in medium and large UMs. However, 23% of patients still experienced tumour recurrence during the follow-up period, and most patients undergoing CKRT suffered from eyesight loss and various radiation-related complications. Since these issues are common among all patients receiving RT, personalized RT plans should be developed based on the patient's overall health, tumour characteristics, and economic capacity in clinical practice.

SUMMARY

What was known before

- Previous clinical studies have suggested that CKRS may be a superior option for treating uveal melanoma, but these studies were mostly smaller, single-arm studies.

What this study adds

- This is the first meta-analysis of CKRS in the treatment of UM.
- The local tumour control rate and eye retention rate after cyberknife radiotherapy are very impressive.
- These findings have important implications for treatment options for uveal melanoma, and because of its advantages of high efficiency, non-invasiveness and low cost, it has great potential for development and application.

DATA AVAILABILITY

The datasets generated and analyzed in this study are available from the corresponding author on reasonable request.

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AUTHOR CONTRIBUTIONS

All authors contributed to the study conception and design. Writing - original draft preparation: Kejie Du; Writing - review and editing: Kejie Du; Conceptualization: Kejie Du; Methodology: Kejie Du; Formal analysis and investigation: Kejie Du; Funding acquisition: Wenjuan Luo; Resources: Kejie Du; Supervision: Wenjuan Luo, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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COMPETING INTERESTS

The authors declare no competing interests.

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